Subject Code: CY1L001	Subject Name: Engineering Chemistry	L-T-P: 3-1-0	Credit: 4

Functional Materials - Design & Application: (a) Synthetic Polymers (carbon framework, silicon framework, fluorinated polymer), Bio & biodegradable polymers. (b) Surfactants. (c) Nanostructures, Soft materials and Thin Films. (b) Emerging applications in Energy harvesting, Memory Storage and Micro-fabrication.
Industrial & Bio-inspired Chemistry: (a) Case studies on Industrial organics with emphasis to Drugs (b) Oxidation, Reduction, Catalytic hydrogenation and Electron transfer.
Molecules in Daily Life: A short tour on molecules behind taste, smell, pain, colour and sex.

Text/Reference Books:

4. Morrison R. T. and Boyd R. N. *Organic Chemistry*, Prentice Hall.
5. Steed J. W. and Atwood J. L. *Supramolecular Chemistry*, John-Wiley.
6. Caruther W. *Reagents in Organic Chemistry*, Cambridge University Press.
7. Wiseman P. *An Introduction to Industrial Organic Chemistry*, Applied Science.
8. Hall N. *The New Chemistry*, Cambridge University Press.

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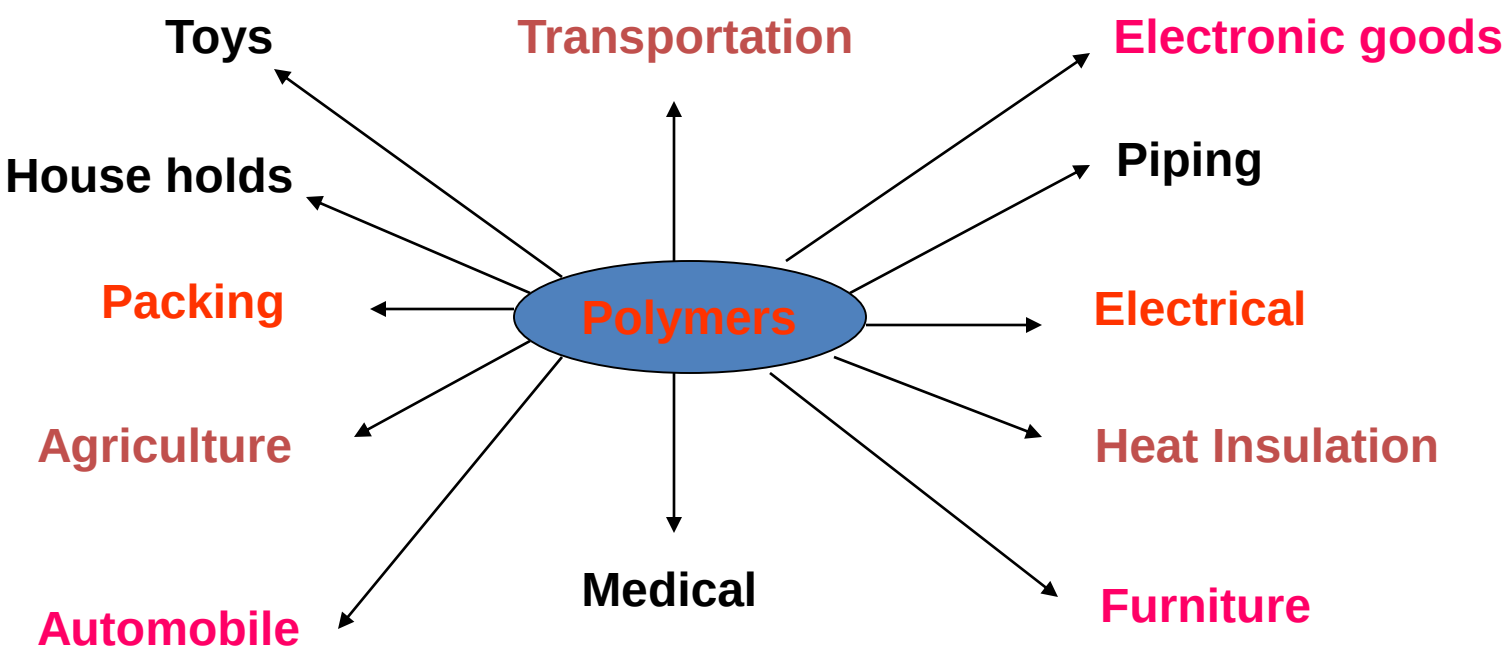
IIT Bhubaneswar

Chemistry (CY01001)

Introduction to Polymers

- Classification of Polymers
- Addition (Chain growth) Polymers
- Condensation (Step growth) Polymers


Toys
Transportation
Electronic goods
House holds
Piping
Packing
Polymers
Electrical
Agriculture
Heat Insulation
Medical
Automobile
Furniture

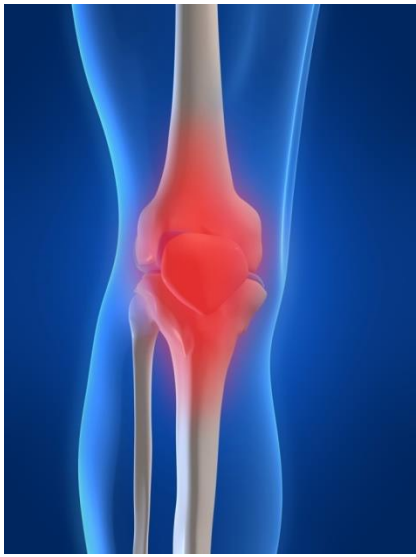
General applications

PET - Soft drinks, water bottles, jam jars
HDPE - water pipes, milk, juice and water bottles
LDPE - frozen food bags, squeezable bottles
PVC - PVC pipes, juice bottle
PP - Reusable microwaveable ware, kitchen ware
PS - Egg carton, disposable cups, plates
PC - baby milk bottles
ABS - baby milk bottles
UPVC - windows

Medical Applications

HDPE - Blood filters, IV fluid bottles, Syringes
LDPE - Urine bags, examination gloves
UHMWHDPE- Joint replacement prosthesies
PP - Membrane supported suture
PVC - Blood bags, Surgical gloves
Polyamides - Artificial Skin
PTFE - Artificial bones and joints

Hip and Knee replacement



In 1958, John Charnley from England introduced the idea of replacing the eroded arthritic socket with a Teflon component. When this failed he used a Ultra High Molecular Weight High Density Polyethylene (UHMWHDPE) polymer to construct the socket.

To obtain fixation of the polyethylene socket as well as the femoral implant to the bone, polymethylmethacrylate (PMMA) was used as a bone cement.

Polymer

Polymers : Poly + mers

Poly means many; **mers** means units or parts

A polymer (macromolecule) is a large molecule built up by the repetition of small, simple chemical units (Monomers)

Polymerization: The fundamental process by which low molecular weight compounds are converted into high molecular weight compounds

$$n\ CH_2 = CH_2 \longrightarrow \text{------}(CH_2\text{-}CH_2\text{-}CH_2\text{-}CH_2\text{-})_n\text{------}$$

ethylene (monomer) polyethylene (polymer)

Monomer

Monomer is the building block or starting material for polymer prepared

Why would a polymer form?

Synthetic polymers:

energy-rich monomers: exothermic reaction

polycondensation: release of small molecule: entropy gain

ring-opening polymerization: release of ring strain

addition polymerization: convert Pi to sigma bond

Types of Polymerization:

Addition and Condensation polymerization [classification is based on the composition or structure of polymers]

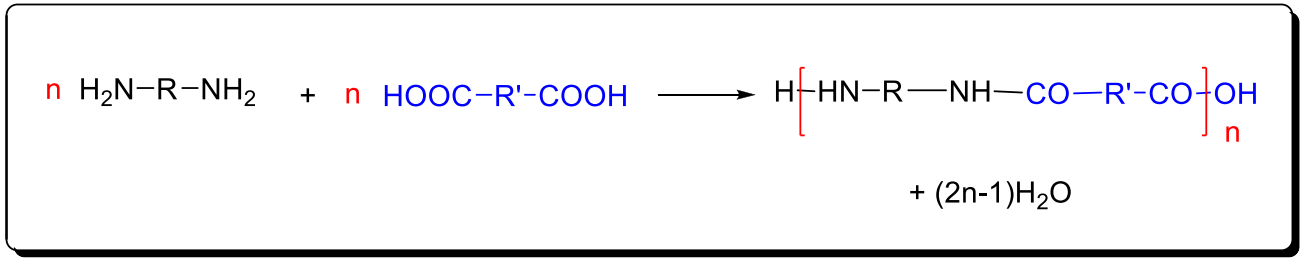
or

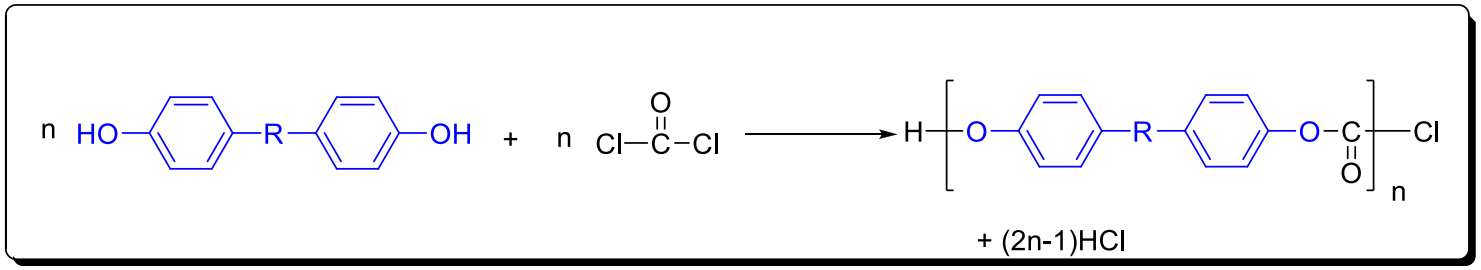
Chain and Step growth polymerization [classification is based on the mechanisms of the polymerization]

Condensation (Step-Growth) Polymerization:

Some of the earliest useful polymeric materials, the Bakelite resins formed from the condensation of phenol and formaldehyde, are examples of step-growth processes.

In general, step-growth polymers such as polyesters and polyamides possess more robust mechanical properties, including toughness, stiffness, and higher temperature resistance, than polymers from addition polymerization processes such as polyolefins and other vinyl-derived polymers.



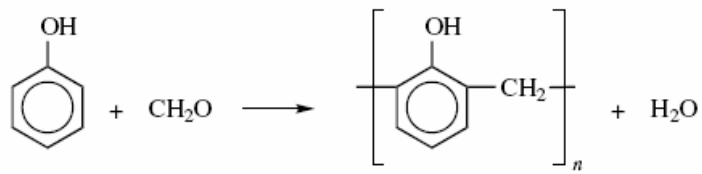
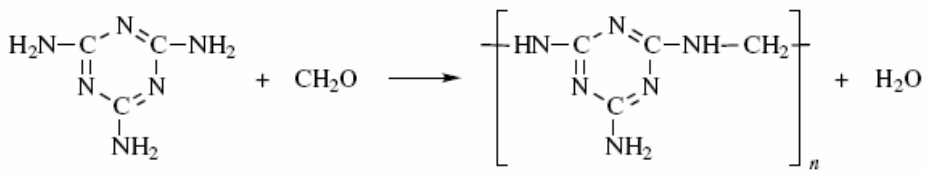


Some naturally occurring polymers such as cellulose, starch, wool, and silk are classified as condensation polymers

Types of Polymerization:

Condensation (Step-Growth) Polymerization:

Typical Condensation Polymerization

Type	Characteristic Linkage	Polymerization Reaction
Polyamide	–NH–CO–	$H_2N-R-NH_2 + HO_2C-R'-CO_2H \longrightarrow H\{NH-R-NHCO-R'-CO\}_nOH + H_2O$
		$H_2N-R-NH_2 + ClCO-R'-COCl \longrightarrow H\{NH-R-NHCO-R'-CO\}_nCl + HCl$
		$H_2N-R-CO_2H \longrightarrow H\{NH-R-CO\}_nOH + H_2O$
Protein, wool, silk	–NH–CO–	Naturally occurring polypeptide polymers; degradable to mixtures of different amino acids.
		$H\{NH-R-CONH-R'-CO\}_nOH + H_2O \longrightarrow H_2N-R-CO_2H + H_2N-R'-CO_2H$
Polyester	–CO–O–	$HO-R-OH + HO_2C-R'-CO_2H \longrightarrow H\{O-R-OCO-R'-CO\}_nOH + H_2O$
		$HO-R-OH + R''O_2C-R'-CO_2R'' \longrightarrow H\{O-R-OCO-R'-CO\}_nOH + R''OH$
		$HO-R-CO_2H \longrightarrow H\{O-R-CO\}_nOH + H_2O$
Polyurethane	–O–CO–NH–	$HO-R-OH + OCN-R'-NCO \longrightarrow \{O-R-OCO-NH-R'-NH-CO\}_n$
Polysiloxane	–Si–O–	$Cl-SiR_2-Cl \xrightarrow[-HCl]{H_2O} HO-SiR_2-OH \longrightarrow H\{O-SiR_2\}_nOH + H_2O$
Phenol–formaldehyde	–Ar–CH_2–	
Urea–formaldehyde	–NH–CH_2–	$H_2N-CO-NH_2 + CH_2O \longrightarrow \{HN-CO-NH-CH_2\}_n + H_2O$
Melamine–formaldehyde	–NH–CH_2–	
Polysulfide	$-S_m-$	$Cl-R-Cl + Na_2S_m \longrightarrow \{S_m-R\}_n + NaCl$
Polyacetal	$-O-\underset{\underset{R}{\vert}}{CH}-O-$	$R-CHO + HO-R'-OH \longrightarrow \{O-R'-OCHR\}_n + H_2O$

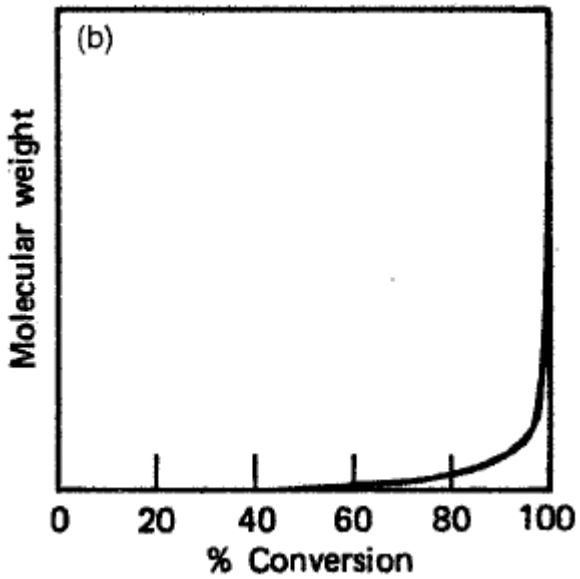
Types of Polymerization:

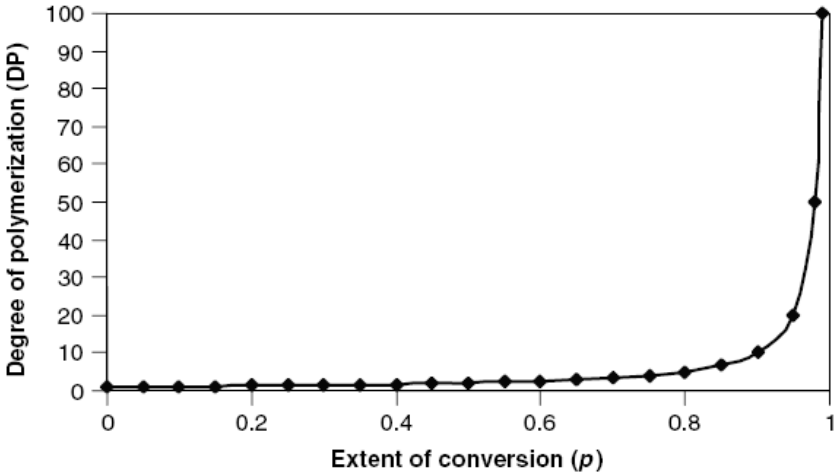
Condensation (Step-Growth) Polymerization:

Monomer + monomer → dimer
Dimer + monomer → trimer
Dimer + dimer → tetramer
Trimer + monomer → tetramer
Trimer + dimer → pentamer
Trimer + trimer → hexamer

→ **Polymer**

Only after 100 % conversion and at the end of the reaction we get maximum molecular weight





Molecular weight vs % conversion in step-growth polymerization

Types of Polymerization:

Addition (Chain-Growth) Polymerization:

- It is a reaction that yields a polymer, which is an exact multiple of the original monomeric molecule.
- The original monomeric molecule, usually, contains one or more double bonds.
- In this addition polymerisation there is no elimination of any molecule.

$$n\,CH_2 = CH_2 \xrightarrow[\text{Catalyst}]{\text{Heat / pressure}} -(CH_2 - CH_2)-_n$$

Ethylene → PE

$$n\,CH_2 = \overset{\overset{Cl}{|}}{CH} \xrightarrow[\text{Catalyst}]{\text{Heat / pressure}} -(CH_2 - \overset{\overset{Cl}{|}}{CH})-_n$$

Vinyl Chloride → PVC

$$n\,CH_2 = \overset{\overset{CN}{|}}{CH} \xrightarrow[\text{Catalyst}]{\text{Heat / pressure}} -(CH_2 - \overset{\overset{CN}{|}}{CH})-_n$$

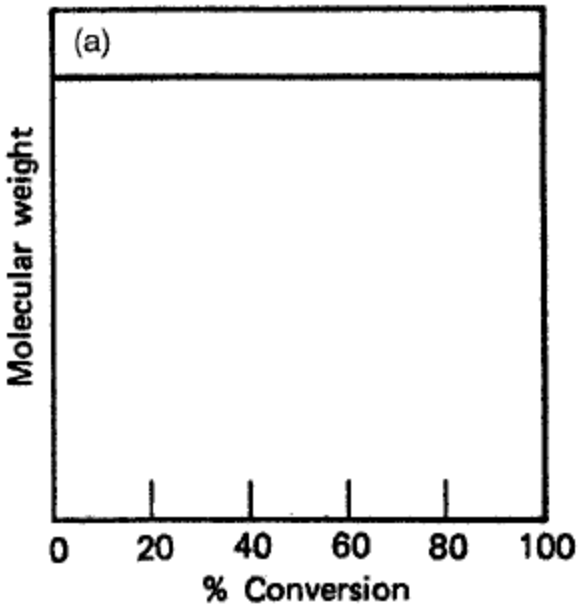
Acrylonitrile → PAN

Addition (Chain-Growth) Polymerization:

Typical Addition Polymerization

Polymer	Monomer	Repeating Unit
Polyethylene	$CH_2{=}CH_2$	$-CH_2-CH_2-$
Polyisobutylene	$CH_2{=}C(CH_3)_2$	$-CH_2-C(CH_3)_2-$
Polyacrylonitrile	$CH_2{=}CH-CN$	$-CH_2-CH(CN)-$
Poly(vinyl chloride)	$CH_2{=}CH-Cl$	$-CH_2-CH(Cl)-$
Polystyrene	$CH_2{=}CH-\phi$	$-CH_2-CH(\phi)-$
Poly(methyl methacrylate)	$CH_2{=}C(CH_3)(CO_2CH_3)$	$-CH_2-C(CH_3)(CO_2CH_3)-$
Poly(vinyl acetate)	$CH_2{=}CH-OCOCH_3$	$-CH_2-CH(OCOCH_3)-$
Poly(vinylidene chloride)	$CH_2{=}CCl_2$	$-CH_2-CCl_2-$
Polytetrafluoroethylene	$CF_2{=}CF_2$	$-CF_2-CF_2-$
Polyisoprene (natural rubber)	$CH_2{=}C(CH_3)-CH{=}CH_2$	$-CH_2-C(CH_3){=}CH-CH_2-$

In Chain-growth polymerization maximum molecular weight will be obtained at the beginning of the reaction irrespective of the monomer conversion as shown in bellow figure



Molecular weight vs % conversion in chain-growth polymerization

Types of Polymerization:

Addition (Chain-Growth) Polymerization:

Essential ingredients

- ✓ Monomer
- ✓ Initiator/catalyst
- ✓ Cocatalyst

Important stages in chain polymerization

1. Initiation:

$$M + I \longrightarrow M^*$$

2. Propagation:

$$M^* + nM \longrightarrow (M)_nM^*$$

3. **Termination / transfer:**

$$(M)_nM^* + AX \longrightarrow (M)_nMX + A$$

1). Free radical polymerization

2). Cationic polymerization

3). Anionic polymerization

Types of Polymerization:

TABLE 3-1 Types of Chain Polymerization Undergone by Various Unsaturated Monomers

	Type of Initiation		
Monomers	Radical	Cationic	Anionic[a]
Ethylene	+	−	+
1-Alkyl alkenes (α-olefins)	−	−	+
1,1-Dialkyl alkenes	−	+	−
1,3-Dienes	+	+	+
Styrene, α-methyl styrene	+	+	+
Halogenated alkenes	+	−	−
Vinyl esters (CH_2=CHOCOR)	+	−	−
Acrylates, methacrylates	+	−	+
Acrylonitrile, methacrylonitrile	+	−	+
Acrylamide, methacrylamide	+	−	+
Vinyl ethers	−	+	−
N-Vinyl carbazole	+	+	−
N-Vinyl pyrrolidone	+	+	−
Aldehydes, ketones	−	+	+

[a]Includes anionic coordination initiation (Chap. 8).

Types of Copolymers

Homopolymer

----AAAAAAAAA-----

----BBBBBBBBB-------

Types of Co-polymers

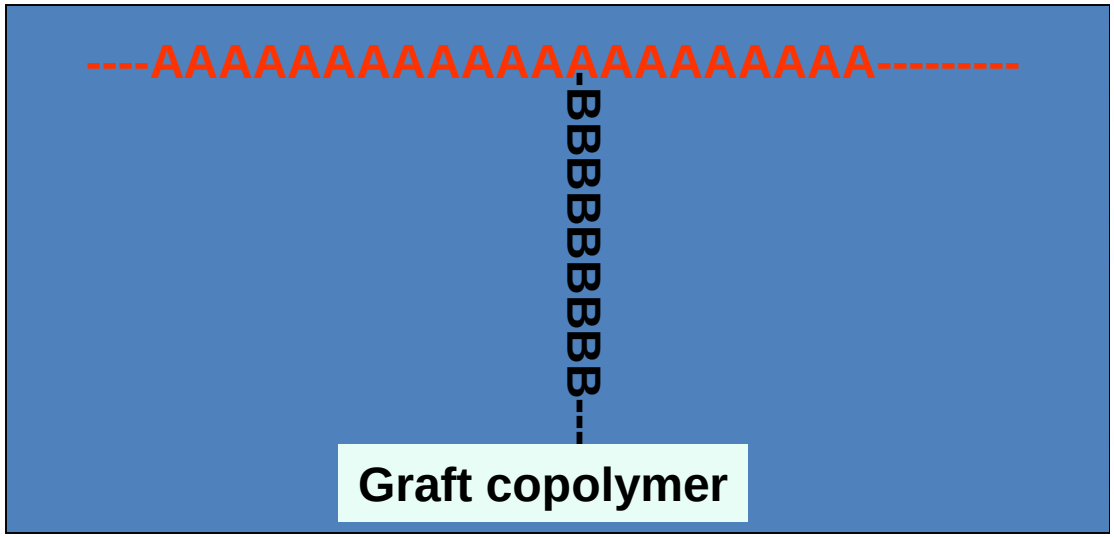
Alternating copolymer
-------ABABABABABABABA-----

Random copolymer
-------AAABABBBAABABBABBA-----

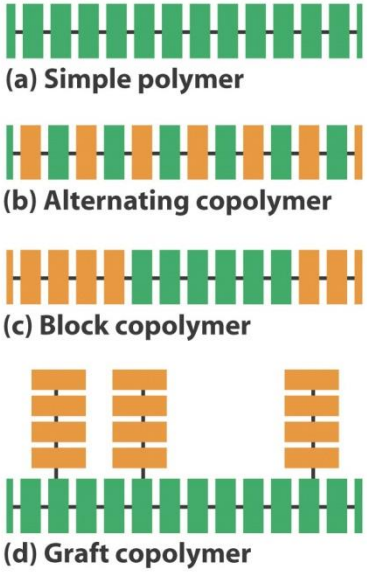
Gradient copolymer
-------AAAABAAABBAABBBABBBB-----

Diblock copolymer
----AAAAA—BBBB---- = A_x-*b*-B_y

Triblock copolymer
--AAAAA--BBBB---CCCCC-- = A_x-*b*-B_y-*b*-C_z



Graft copolymer



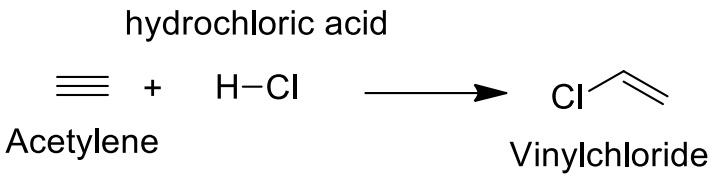
Synthesis of Addition Polymers

Polyvinyl Chloride (PVC)

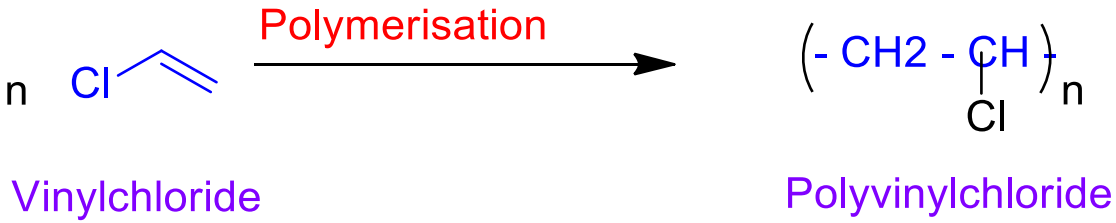
Polyvinyl Chloride (PVC)

Preparation: Preparation of PVC involves the following two steps

I step: Vinyl chloride is prepared by treating acetylene with hydrogen chloride at 60-80°C in the presence of metal chloride at catalyst



II step: Polyvinylchloride is obtained by heating water emulsion of vinyl chloride in presence of benzoyl peroxide (or) hydrogen peroxide under pressure



Polyvinyl Chloride (PVC)

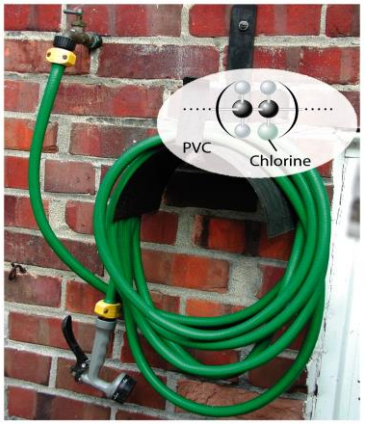
Properties of PVC:

1. PVC is colourless, odourless and chemically inert powder
2. It is insoluble in inorganic acids and alkalis, but soluble in hot chlorinated hydrocarbons such as ethylchloride
3. It undergoes degradation in presence of heat (or) light

Uses of PVC

1. It is used in the production of pipes, cable insulations, table covers and rain-coats etc.,
2. It is also used for making sheets, which are employed for tank-linings, light fittings, refrigerator components, etc.,

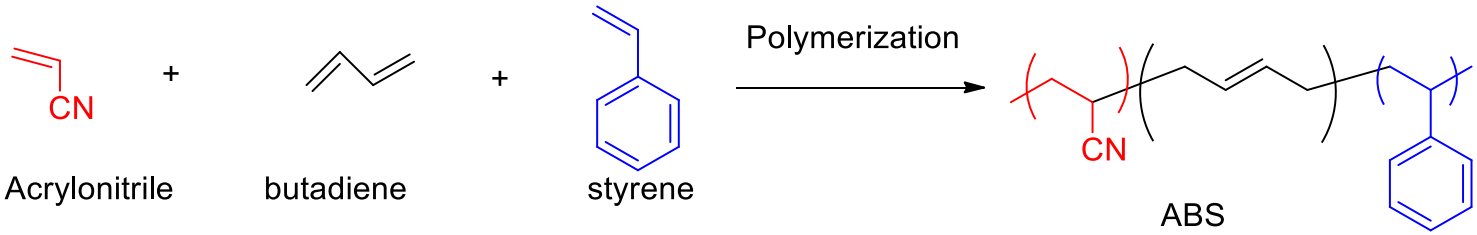
Polyvinylchloride (PVC)



Acrylonitrile Butadiene Styrene(ABS)

ABS is a ter-polymer

It is polymerized by the polymerization of acrylonnitrile, butadiene and styrene



It is an amorphous and thermoplastic polymer

Mainly ABS can be prepared in two different grades: 1. for extrusion and 2. for injection moulding.

Acrylonitrile Butadiene Styrene(ABS)

Properties of ABS:

- Cost effective
- Good Impact Strength
- Good Chemical Resistance
- High Gloss Surface Finish
- Good Flexural Properties
- Can prepare in different grades as per the requirement

ABS can be recycled

Applications of ABS:

- musical instruments (recorders, plastic clarinets, and piano movements)
- automotive trim components, automotive bumper bars
- medical devices for blood access
- White-water canoes
- buffer edging for furniture and joinery panels
- luggage and protective carrying cases
- small kitchen appliances
- Keyboard keycaps
- toys, including Lego and Kre-O bricks
- golf club heads (because of its good shock absorbance)
- Household and consumer goods

Acrylonitrile Butadiene Styrene(ABS)

- **ADVANTAGES:**
- Good impact resistance with toughness and rigidity
- Metal coatings have excellent adhesion to ABS
- Formed by conventional thermoplastic methods
- A light-weight plastic

DISADVANTAGES AND LIMITATIONS:

- Poor solvent resistance
- Low dielectric strength
- Only low elongations available
- Low continuous service temperature

TYPICAL ENGINEERING APPLICATIONS:

- Automotive hardware (used in electroplated metal coatings for decorative hardware), appliance cases, pipe, plated items.

Synthesis of Condensation Polymers

Bakelite

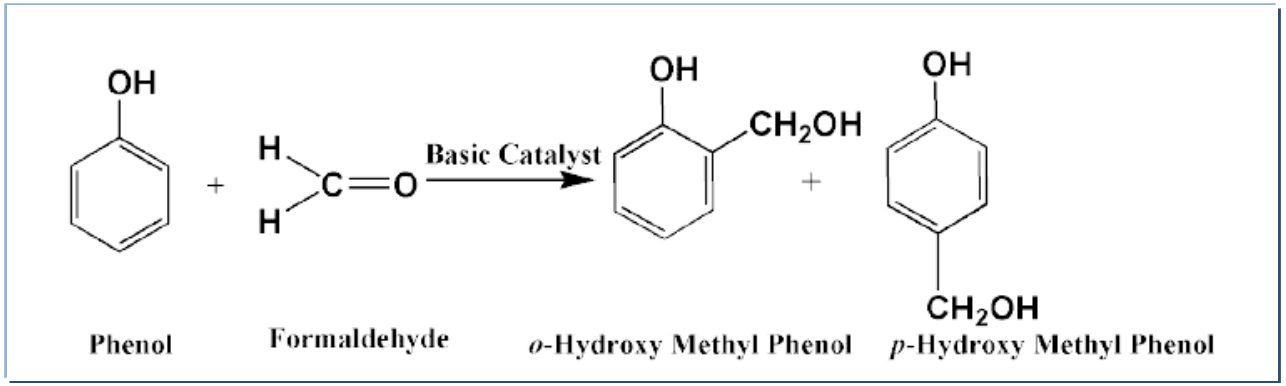
Bekalite is obtained by the condensation polymerisation of phenol and formaldehyde in the presence of acid or alkali catalyst

Preparation

The reaction involves the following 3 steps

Stage:1

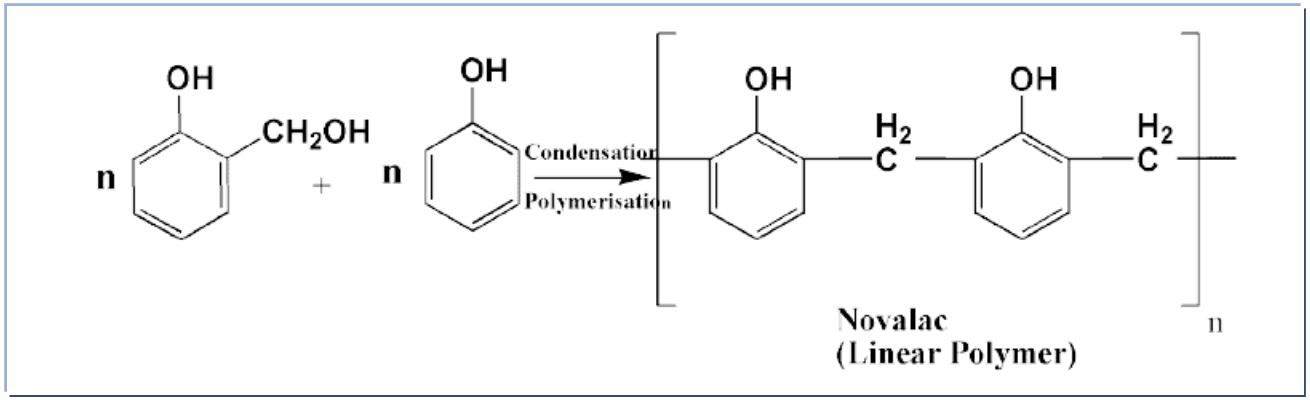
The first step is the reaction between phenol and formaldehyde to produce methylol phenols (mono, di and tri-methylol phenols)



Bakelite

Stage 2: Synthesis of linear polymer Novolac

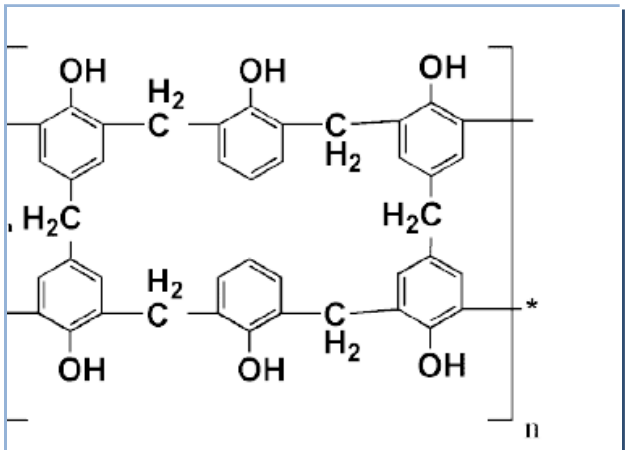
- When methylol phenols are heated with excess of formaldehyde in presence of alkaline catalyst. The methylol phenols condense either through methylene linkages or through ether linkages to form resoles.
- **Resole** is a low molecular weight linear polymer. It is completely soluble in alkaline solution



Bakelite

Stage 3: Synthesis of cross linked polymer Bakelite or phenol formaldehyde resin

The above Novalac upon heating further in presence of HCHO in the presence of a curing agent (hexamethylene tetramine) produces hard, rigid, infusible, cross-linked polymer called bakelite



Bakelite
(Cross-linked Polymer)

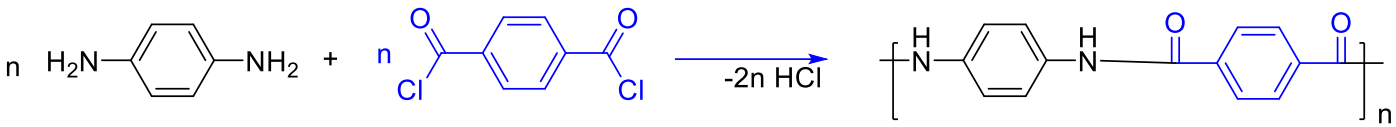
Bakelite

- **Properties of Bakelite**
- Bakelite is resistant to acids, salts and most organic solvents, but it is attacked by alkalis because of the presence of –OH groups
- It possesses excellent electrical insulating property

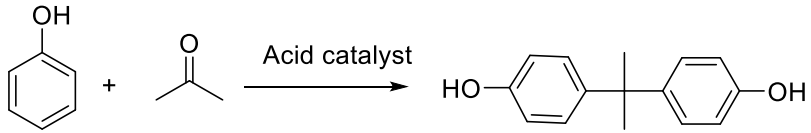
- **Uses of Bakelite**
- Bakelite is used as an adhesive in plywood laminations & grinding wheels, etc
- It is also widely used in paints, varnishes
- Cooker with **Bakelite** Handles
- It is used for making electrical insulator parts like plugs, switches, heater handles, etc.,

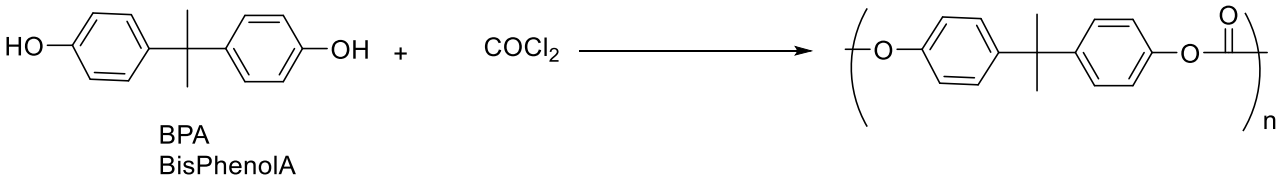
Kevlar

Poly-paraphenylene terephthalamide



Polycarbonates





Classification of Polymers

Classification of Polymers

1. **Classification based on the Source:**
   (a). Natural and (b). Synthetic Polymers
2. **Classification based on Structure:**
   (a). Linear (b). Branched and (c). 3D Network
3. **Classification based on method of synthesis:**
   (a). Addition Polymers and (c). Condensation Polymers
4. **Classification based on growth of polymer chain (Mechanism)**
   (a). Chain growth Polymerization and (b). Step growth Polymerization
5. **Classification based on Molecular forces**
   (a). Thermoplastic (b). Thermosetting (c). Elastomers (d). Fibers
6. **Classification based on Nature:**
   (a). Degradable and (b). Non-degradable
   (i).Bio-degradable (promoted by enzymes)
   (ii). Photodegradable
   (iii). Environmental Erosion (Wind, rain, temperature etc weather elements
   (iv). Chemical degradation

Thermoplastic and Theromosetting

Polymers which can be softened on heating and harden on cooling reversibly, are called thermoplastics. Repeated heating and cooling do not alter the chemical nature of these materials, because the changes involved are purely of physical nature.

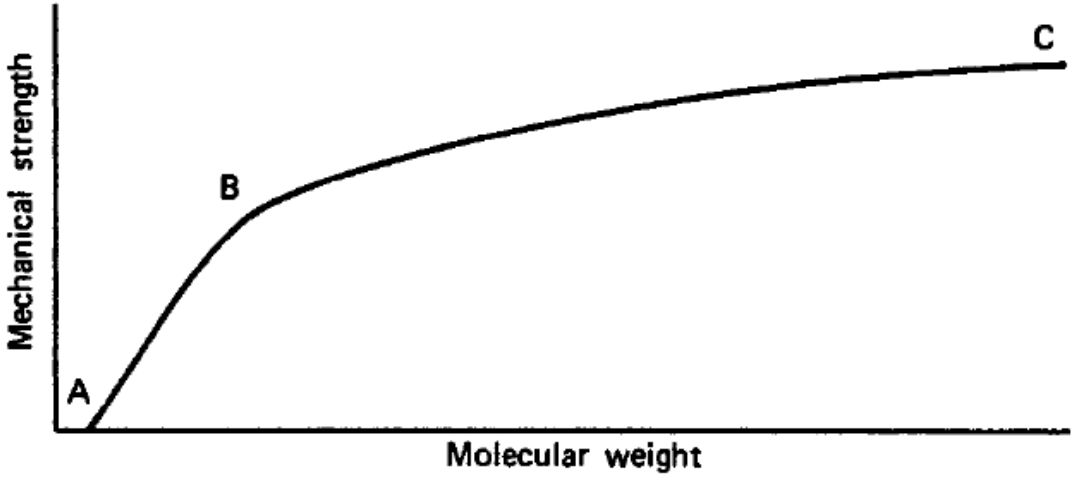
Eg:. polyethylene, polypropylene, poly(vinyl chloride), polystyrene, polytetrafluro ethylene (PTFE or Teflon) etc**.**

Polymers which on heating get hardened and once they have solidified, they cannot be softened, i.e. they are permanent setting resins are called thermosetting plastics.

Eg. polyesters, Bakelite and epoxy resins etc.

Molar Mass of the Polymer

Molar Mass of the Polymer



Dependence of mechanical strength on polymer molecular weight

A: minimum polymer molar mass, usually a thousand or so, to produce any significant mechanical strength at all.
B Critical Molar Mass
C Limiting value

Polymer molecular weight

Number Average Molecular Weight (M_w) =

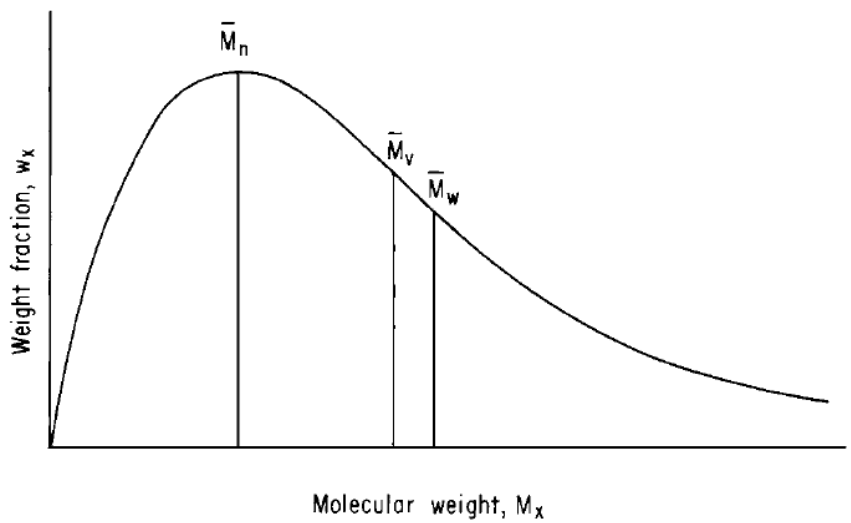
$$\overline{M}_n = \frac{w}{\sum N_x} = \frac{\sum N_x M_x}{\sum N_x} = \sum \underline{N}_x M_x$$

Weight average Molecular Weight (M_w) :

$$\overline{M}_w = \sum w_x M_x = \frac{\sum c_x M_x}{\sum c_x} = \frac{\sum c_x M_x}{c} = \frac{\sum N_x M_x^2}{\sum N_x M_x}$$

Viscosity-average molecular weights:

$$\overline{M}_v = \left[\sum w_x M_x^a\right]^{1/a} = \left[\frac{\sum N_x M_x^{a+1}}{\sum N_x M_x}\right]^{1/a}$$



Distribution of molecular weight for polymer sample

Various methods based on solution properties are used to determine the average molecular weight of a polymer sample. These include methods based on 1. colligative properties, 2. light scattering, and 3. viscosity

Methods of measuring molecular weights:

1. Colligative properties of solutions: (Absolute method, which don't need knowledge of physical and chemical structure of polymer molecules)

—vapor pressure lowering (vapor pressure osmometry), freezing point depression (cryoscopy), boiling point elevation (ebulliometry), and osmotic pressure (membrane osmometry).

(a) ***Osmometry*****.** This is a vapour pressure method, useful for polymers of molecular weight up to about 25 000; membrane osmometry is used for molecular weights from 20 000 to 1 000 000. These are number average methods.

2. ***Viscometry*****.** This is a **relative method**, but the simplest, and its application is widespread in industry. It gives viscosity average molecular weight (Mv). Viscometry is approximately a weight average method.

3. ***Light scattering*****. (Absolute method)**Light scattering by polymer solutions, unlike coligative properties, is greater for larger-sized molecules than for smaller-sized molecules. Thus Light Scattering is a weight average method. (it measures Mw)

4. ***Gel permeation chromatography.*** This is a **direct fractionation method** using molecular weight. It is relatively rapid and has proved to be one of the most valuable modern methods.

5. ***Chemical methods.*** These usually depend on measuring distinctive end groups. They are number average methods.

Encountered in Polymer Chemistry

Average	Definition	Methods
$\bar{M}_n$	$\frac{\Sigma_i N_i M_i}{\Sigma_i N_i}$	Osmotic pressure and other colligative properties End group analysis
$\bar{M}_w$	$\frac{\Sigma_i N_i M_i^2}{\Sigma_i N_i M_i}$	Light scattering Sedimentation velocity
M_z	$\frac{\Sigma_i N_i M_i^3}{\Sigma_i N_i M_i^2}$	Sedimentation equilibrium
$\bar{M}_v$	$\left(\frac{\Sigma_i N_i M_i^{1+a}}{\Sigma_i N_i M_i}\right)^{1/a}$	Intrinsic viscosity

"a" is a constant it depends upon hydrodynamic radius of the polymer chains in the given solvent. Its value is between 0.5-0.9.

Absolute methods include colligative methods (membrane osmometry, ebulliometry, cryoscopy, and vapor-phase osmometry), light scattering, and equilibrium sedimentation.